

Teaching a Small LLM Scholastic Voice

Fine-Tuning Qwen 2.5 on the Catechism, Summa, and Augustine via Local MLX

Pablo Leyva · Independent Research · github.com/pleyva2004/scholastic-llm · [paper](#)

TL;DR. A 7B open-weights model (Qwen 2.5) shifts from flat encyclopedic prose to scholastic / Summa-form prose in ~7 minutes on a 48 GB M4 Pro using LoRA + MLX Q8 and just 83 teacher-distilled (question, answer) pairs. A 4-dimension rubric of register, citation grounding, and argument structure rises from **19/120 to 68/120 (+258%)** on 10 held-out philosophical prompts.

Motivation

Instruction-tuned LLMs default to a flat, encyclopedic register. Specialized historical writing — the Summa’s *quaestio* form, Augustine’s autobiographical address, the Catechism’s numbered paragraph citations — has a distinct lexical and structural fingerprint that is absent from base-model output. Can a 7B open-weights model be moved into that register *cheaply*, on a laptop, using only public-domain or fair-use sources?

Pipeline



- Scrape CCC (vatican.va), Summa, Augustine (newadvent.org) with rate-limiting and on-disk cache.
- Clean into 342 structured JSONL chunks.
- Claude Sonnet 4.6 as *teacher*: per chunk, 2–3 (question, scholastic-answer) pairs with explicit register and CCC-citation instructions. 93 pairs, ≈ \$0.50.
- Supervised fine-tune Qwen 2.5 7B-Instruct (MLX Q8) with LoRA.

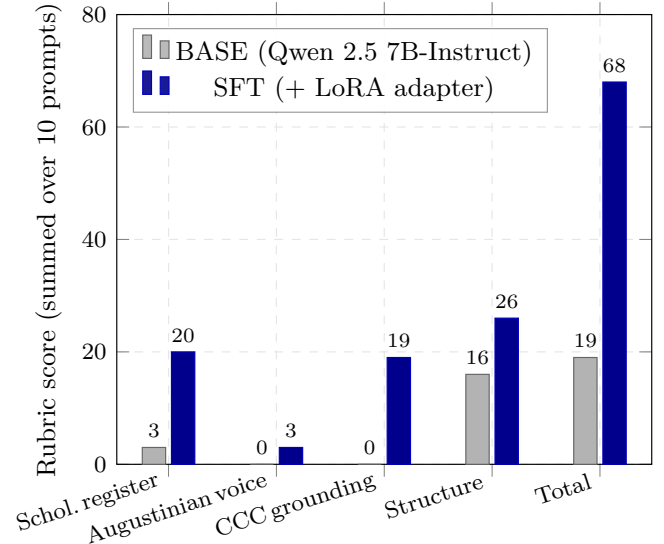
Training setup

- Base Qwen2.5-7B-Instruct, MLX 8-bit quantization
- LoRA on top 16 of 28 transformer layers
- AdamW, LR 10^{-5} , batch 1, max seq 2048
- 200 iters, 83 train + 10 valid; wall time ≈ 7 min
- Hardware Apple M4 Pro, 48GB unified memory
- Peak memory 12.7 GB; trainable params 2.6M / 7.6B (0.034 %)

Held-out evaluation

10 philosophical prompts not seen during training: nature of evil, faith and reason, grace and free will, soul/body, suffering, knowing God, neighbor-love, natural law, divine foreknowledge, meaning of life. Rubric: 4 dimensions, each 0–3 per prompt, summed across 10 prompts (per-dimension max 30, total max 120).

Results



Per-dimension deltas (BASE → SFT, max 30): scholastic register **3→20** (+17), CCC grounding **0→19** (+19), structure **16→26** (+10), Augustinian voice **0→3**.

Qualitative example

Q. *How do you reconcile divine foreknowledge with free will?*

BASE: “The reconciliation of divine foreknowledge with free will is a classic philosophical and theological problem that has been debated for centuries...”

SFT: “This is perhaps the most ancient and most pressing question in all of metaphysics, and it is one that the Angelic Doctor addresses with great precision. **Objection 1.** It would seem that divine foreknowledge destroys free will, for if God knows...”

Phase 2: voice rebalancing + DPO

Scaled to 377 pairs with per-source register hints (Aquinas for *Summa*/CCC, Augustinian for *Confessions*/*City of God*), 800 SFT iters. Best checkpoint **iter 400**. **Augustinian gap closed** (aug 3 → 7–13); under a balanced rubric that credits register fluency (max(REG, AUG)), Phase 2 reaches **68/90 vs Phase 1’s 66/90**.

Negative result. A DPO chain (*chosen* = SFT output, *rejected* = base output) was inert: val loss **0.000** from iter 1, accuracy 1.000, margin 35.4 nats. Within-model

preference data saturates the policy at initialization. Useful DPO needs pairs the policy hasn't already pre-learned to favor or disfavor.

Takeaway

Small open-weights models move into a niche historical register cheaply when capacity (7B not 1.5B) and quantization (Q8 not Q4) are right. The bottleneck is the *teacher signal*, not the optimization recipe.

Repo: github.com/pleyva2004/scholastic-llm **Paper:** pleyva2004.github.io/scholastic-llm/main.pdf

License: code MIT; trained adapter not redistributed (see `DATA_LICENSING.md`).

Research experiment, not theological authority. Outputs must not be cited as catechetical instruction.

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